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Email Submission: [REDACTED] Document Number: 2025-02305

**Re: Request for Information on the Development of an Artificial Intelligence (AI) Action Plan<sup>1</sup>**

Dear Mr. D'Souza:

Lockheed Martin Corporation (Lockheed Martin, LM) appreciates the opportunity to submit these comments in response to the Networking and Information Technology Research and Development (NITRD) National Coordination Office's (NCO), National Science Foundation (NSF) Request for Information (RFI), *Development of an Artificial Intelligence (AI) Action Plan*.<sup>2</sup> Lockheed Martin commends the White House for its strong AI leadership, as evidenced by *Executive Order 14179: Removing Barriers to American Leadership in Artificial Intelligence*.<sup>3</sup> Undoubtably, the U.S. must sustain and enhance its significant global AI leadership to promote economic competitiveness, national security, and human flourishing—the AI Action Plan will play a critical role in achieving this objective.

Lockheed Martin is a global enterprise principally engaged in the research, design, development, manufacture, and integration of next-generation technologies, systems, products, and services for both government and commercial customers worldwide—many of which are seeking to leverage AI. One such Lockheed Martin example is the development of AI agents for the Defense Advanced Research Projects Agency's (DARPA) AlphaDogfight trials, a three-day competition designed to demonstrate advanced algorithms capable of performing

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<sup>2</sup> NSF, *Request for Information on the Development of an Artificial Intelligence (AI) Action Plan* (rel. Feb. 6, 2025).

<sup>3</sup> EO 14179: *Removing Barriers to American Leadership in Artificial Intelligence* (Jan. 23, 2025).

simulated, within-visual-range air combat maneuvering.<sup>4</sup> We also recognize the important role that industry partnerships will play in furthering the science and application of both AI and machine learning (ML). For example, Lockheed Martin has engaged in strategic partnerships covering a breadth of infrastructure needs with companies such as NVIDIA, Microsoft, and Meta; and, through joint technology agreements with companies such as Intel and IBM, is working to advance the science of state-of-the-art technologies, such as neuromorphic computing—where elements of a computer seek to mimic the operation of a human brain.

Given the significant battlespace role potential for AI, Lockheed Martin has also made enduring investments in its own in-house AI capabilities, including establishing the AI Center,<sup>5</sup> and a subsidiary company, Astris AI.<sup>6</sup> These investments have yielded significant results, including the deployment of hundreds of large language model applications, processing over 3,000,000,000 tokens per week. Appendix I to these comments presents an LM-authored Institute of Electrical and Electronics Engineers (IEEE) paper highlighting these successes.

Lockheed Martin offers the following for consideration as part of the AI Action Plan:

### **National Security and Defense**

For AI in national security applications, Lockheed Martin believes a “human-in-the-loop” is always necessary to ensure human judgement and oversight of a system’s decisions, regardless of other nations’ stated positions on the matter. Additionally, and as discussed in the *Explainability and Assurance of AI Model Outputs* section below, decision traceability requirements should similarly be more stringent for national security applications—Lockheed Martin has similarly endorsed this principle as part of the Defense Innovation Board’s principles for the ethical use of AI.

The United States Government (USG) should also consider further investing in edge-AI technologies and low-power systems for use in the field—examples are those developed by Lockheed Martin under DARPA’s Artificial Intelligence Reinforcements contract. Edge-AI is already yielding significant benefits to our military: Lockheed Martin has successfully made AI/ML updates to the Navy’s Aegis Combat System improving how Navy destroyers counter Houthi missile and drone attacks in the Red Sea.<sup>7</sup>

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<sup>4</sup> See <https://www.darpa.mil/news-events/2020-08-26>.

<sup>5</sup> See <https://www.lockheedmartin.com/en-us/capabilities/artificial-intelligence-machine-learning.html>.

<sup>6</sup> For more on Astris AI, see <https://astrisai.com/>.

<sup>7</sup> See <https://www.defensenews.com/naval/2024/03/21/us-navy-making-aegis-updates-training-changes-based-on-houthi-attacks/>.

Edge processing provides servicemembers with critical advantages during combat operations by leveraging the Department of Defense's (DoD) vast data streams to develop actionable intelligence, without having to upload significant amounts of raw data to the cloud; and similarly, low-power systems could prolong the use of an AI capability in the field during persistent operations without requiring the transport and carry of additional energy resources.

Finally, as the battlespace will not be segregated into individual silos based upon security classifications, AI training environments cannot be compartmentalized. Those few settings designed for mixed classification use are not properly configured to support AI model training. Lockheed Martin recommends considering whether AI-specific security classification reform is necessary to enable the multi-classification training environment which most accurately mimics the real world.

### **Regulation and Governance**

Clear regulatory guidelines can be enabling to any fast-paced, burgeoning industry, and AI is no exception. Generally, Lockheed Martin is supportive of AI policy and regulation that clearly establishes guardrails for acceptable behaviors. However, and echoing Vice President Vance's AI Action Summit speech,<sup>8</sup> LM cautions against an overly burdensome regulatory regime which may inadvertently stifle innovation—thus placing U.S. AI leadership on precarious footing. A strong USG-industry partnership will be able to identify regulatory roadblocks to innovation, overlaps, and gaps. We applaud the Administration's efforts to-date to create a business- and innovation-friendly regulatory environment while also recognizing the need for public discussion.

A federal regulatory and governance regime is critical to ensuring that innovators are not subject to a patchwork of inconsistent state-level requirements. Some such proposals mimic the European Union's more restrictive approach to AI. Given President Trump's desire to promote innovation and focus on deregulation, federal AI governance leadership may seek to minimize the potential for significant differences among states, which can impose additional costs and burdens.

There exists a vast array of AI models and capabilities, many of which are tailored to specific tasks—regulatory regimes should be similarly constructed. Specifically, it is important to clearly differentiate between AI as a component of a national security system, and what many legislative proposals term “high-risk AI use cases”, *i.e.*, AI used in making decisions with materially legal or otherwise significant impacts to consumers (*e.g.*, access to lending services, insurance, etc.). Several draft federal and state proposals have resulted in confusion regarding

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<sup>8</sup> Remarks by Vice President Vance at the AI Action Summit (Feb. 11, 2025).

whether national security systems are within scope—often because these proposals rely on multiple preexisting, conflicting terms. As such, it may also be necessary to create new definitions for use in the AI context to best ensure an intended outcome is achieved.

For any federal AI legislation/regulation, it would also be helpful to create an exemption process for AI developed exclusively for national security use cases or integration into a national security system as it relates to “high-risk AI use cases.” It may likely also be necessary to develop a complementary set of requirements exclusive to national security AI.

### **Hardware and Chips**

Access to the necessary chips and hardware is critical to the success of the U.S. AI industry, as well as mitigating associated national security vulnerabilities—such as supply chain disruptions due to any number of factors. We emphasize the imperative of, and strongly recommend that, the USG substantially invest in domestic semiconductor capabilities—both manufacturing infrastructure and expertise. Such investment will help to guarantee the U.S.’ chip supply, providing a critical competitive advantage while also increasing supply chain resilience and shoring up the associated national security concerns. Lockheed Martin notes the implementation of the CHIPS Act is expected to play a key role in strengthening domestic capabilities.

With respect to national security applications for AI specifically, we also believe it is important to ensure that access to chips and other necessary technologies be guaranteed to the DoD and the defense industrial base (DIB). Lacking guaranteed access, the DIB’s small scale may not be able to compete with commercial demand during a supply crisis.

### **Data Centers and Energy**

Training and developing AI systems and ML models requires significant amounts of data storage and computing power. While data centers are well-equipped to meet these needs, they consume vast quantities of electricity and water, with one Department of Energy (DOE) study predicting data centers will consume 6.7-12% of total U.S. electricity by 2028.<sup>9</sup> To address this demand, the U.S. should stimulate investment in the research & development of baseload generation infrastructure, such as small modular nuclear reactors; smart grid technologies that can better match electric supply and demand in real-time; and improved energy storage technologies, *e.g.*, batteries, which would increase data center resilience.

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<sup>9</sup> DOE, *DOE Releases New Report Evaluating Increase in Electricity Demand from Data Centers* (Dec. 20, 2024), <https://www.energy.gov/articles/doe-releases-new-report-evaluating-increase-electricity-demand-data-centers>.

Resilience will also be essential for electric substations and transmission systems. For instance, large power transformers take months to construct and have long lead times—if one such substation transformer servicing a data center were to go down, the center’s power supply may be disrupted for months absent power being supplied from a secondary source or a replacement transformer being readily available.

Increased energy demand will require that data centers themselves seek to adopt new and emerging technologies to achieve greater efficiency. Example technologies include specialized semiconductors for data centers and AI training use cases, as current graphics processing unit-based capabilities are costly in terms of energy consumption and waste heat generation; and brain-inspired, neuromorphic computers, which can be less energy intensive than current data center systems.

Streamlined regulations for permitting, access to energy, and environmental processes are essential to the U.S.’ ability to lead the world in AI; multi-year regulatory processes slow down American companies, and can both drive offshoring to more permissive regulatory countries and provide other nations with valuable time to catch up to the U.S.

### **Model Development and Risk**

AI will continue to gain in prevalence, and so too will the associated risks, thus we recommend developing standardized risk classifications and a use-case specific risk identification framework—as risk(s) will depend greatly upon an AI model’s intended use. Further, and in coordination with industry and academia, we believe the Department of Commerce’s (DOC) National Institute of Standards and Technology (NIST) is uniquely suited to develop an AI risk management framework (RMF), which is essential for the development of traceable, trustable AI. NIST’s previous successes in developing cybersecurity standards demonstrate its capacity to develop an AI RMF.

We also recommend enhancing existing standards and regulations for incident reporting, to include high-risk, high-impact AI-related incidents; creating sector-specific assurance mechanisms for critical infrastructure sectors; and developing an authoritative database of trusted AI models, given concerns related to open-source model provenance.

National security AI use cases are significantly different than for other applications. While we support a national security exemption for “high-risk AI use cases”, as discussed in the above *Regulation and Governance* section, AI for national security should not be exempted from an RMF. The DoD and the DIB should collaborate to identify the appropriate risk tolerances and a national-security specific RMF. While these risk tolerances should flow top-down, there should also be an appropriate amount of flexibility to adjust to program-specific needs.

## **Open-Source Development**

Lockheed Martin believes an AI Bill of Materials (AIBOM) is key to building safe, trustworthy AI. Beyond the above discussion regarding provenance, open-source models present unique challenges. For instance, it is challenging to determine whether open-source models are compliant with AI principles, and they often lack supply chain transparency and validation. This tacitly shifts the burden to consumers, who are unequipped to judge model “goodness”. An AIBOM would greatly increase consumer confidence in open-source models and would be even more valuable for DIB systems, which represent prime targets for adverse action by state and non-state actors alike. An AIBOM is not dissimilar from a software bill of materials, which the Cybersecurity & Infrastructure Security Agency (CISA) has recognized as “a key building block in software security and software supply chain risk management”,<sup>10</sup> with an AIBOM’s additional requirements helping to address the greater complexity and opacity associated with AI models.

Standards such as digital signatures for AI models and provenance tracking would also assist in assessing open-source models. Tamper-proof digital signatures could prove especially useful. If a model is altered without access to the cryptographic signing credential held by the true author or publisher, the signatures will no longer match—indicating an (potentially malicious) alteration to the model. Further, the widespread acceptance of these techniques could themselves provide an additional layer of security, as consumers are likely to look upon unsigned (or otherwise unverified) models with additional scrutiny, as verification would be the expected norm.

We also believe fostering the development of open-source models is critical to enabling a competitive development environment and will further assist in system interoperability. It would be detrimental to U.S. leadership for AI to be developed primarily in proprietary ecosystems. Likewise, encouraging modular, interoperable AI solutions will help customers—including federal agencies—avoid vendor lock-in.

## **Explainability and Assurance of AI Model Outputs**

The absence of regulatorily-standardized explainability requirements has slowed the adoption of AI, especially as it relates to “high-risk” models, in regulated industries, such as the financial sector. Explainability is critical for assessing model accuracy, fairness, and transparency, which in turn impacts trust and confidence in a given model. It’s also unlikely all use cases and models will require the same level of explainability. Lockheed Martin recommends explainability requirements (*e.g.*, what constitutes sufficient explainability) be based upon the above-discussed risk classifications, and further that safety of life and other critical systems be required to display higher degrees of traceability.

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<sup>10</sup> CISA, Software Bill of Materials (accessed Mar. 5, 2024) <https://www.cisa.gov/sbom>.

## Cybersecurity

AI has the potential to address the escalating speed, complexity, and frequency of cyber threats, providing a powerful new tool against malign cyber actors. High-risk AI models, however, are themselves targets for cyber-attack and are vulnerable to unique cyber threats. We recommend the development of cybersecurity standards to counter AI-specific attacks, including, but not limited to: model inversion, where the output of a model is used to infer something about a model's parameters or architecture, *i.e.*, the non-public characteristics of a system; prompt injection, which, similar to traditional Structured Query Language injection attacks, obfuscate malicious prompts to manipulate an AI system into ignoring its guardrails, *e.g.*, leaking sensitive data; and data poisoning, where the training dataset of an AI model is intentionally compromised to influence or manipulate the operation of that model.

As discussed in *Model Development and Risk*, we believe NIST's solid track record on developing cybersecurity standards makes it a logical choice for developing standards to address AI-specific cyber threats. LM also recommends leveraging Federally Funded Research and Development Centers (FFRDC) given their instrumentality in developing AI cybersecurity norms based upon cyber principles.<sup>11</sup>

## Conclusion

Lockheed Martin looks forward to working with the Administration, on the development of the AI Action Plan, and, more generally, on ensuring that the United States remains the undisputed global leader in AI.

Sincerely,

**LOCKHEED MARTIN CORPORATION**

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<sup>11</sup> See, e.g., <https://www.mitre.org/news-insights/publication/principles-reducing-ai-cyber-risk-critical-infrastructure-prioritization> (Oct. 25, 2023).

## Appendix I



# Lockheed Martin AI Factory

## *Generative AI and MLOps for Engineering, Enterprise and Edge*

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**Abstract**—This article reports on the rapid creation and deployment at scale of hundreds of Large Language Model (LLM) applications, highlighting several across enterprise, engineering and edge use cases. This outcome was accelerated by the creation of an open architecture, secure and scalable generative AI Factory, a platform that empowers thousands of developers and over 50,000 end users across a diverse set of data types and use cases throughout our global enterprise. The approach reveals how to affordably deploy generative AI to create value securely at scale.

**Keywords**—Generative AI, AI Factory, LLMs, scale, security

### I. INTRODUCTION

Generative AI promises significant value creation driven by improvements in innovation, productivity, and cost avoidance. Illustrative task acceleration benefits include 29% faster programming for novices, 25% faster consulting, and 34% faster customer service [1]. Automation and productivity opportunities leveraging LLMs are present in all domains and analysts project \$2.6-\$4.4 trillion in value creation potential across 66 use cases [2]. In spite of this exciting potential, most organizations are stuck in pilot purgatory, stalled by unclear outcomes, limited LLM access, risk aversion, and/or inability to scale. After describing a range of dozens of operational use cases at Lockheed Martin, this article exemplifies use cases deployed responsibly and at scale in a new distributed machine learning operations and generative AI platform which currently supports over 10,000 engineers and developers and 50,000 LLM users, processing over 3B tokens per week. Given continuous evolution, we introduce the Astris AI<sup>TM</sup> platform as the next phase of the Lockheed Martin AI Factory<sup>TM</sup>.

### II. GENERATIVE AI USE CASES

Lockheed Martin's hundreds of use cases can be described by their purpose, be it for enhanced engineering, enterprise operations, or edge customer missions. Figure 1 illustrates some of the broad range of use cases in operation at Lockheed Martin used within and across businesses in air, space, missiles, and rotary and mission systems. We illustrate these classes in turn.



Fig. 1. Example use cases

#### A. Engineering Use Cases

Large language models are effective not only at processing (summarizing, paraphrasing, or translating) natural language, but equally at transforming formal languages such as genetic code, computer code, and mathematical formulas or logic. For example, as shown on left side of Figure 2, LM engineers enhance their productivity [3] as well as code quality (efficiency, readability, security) by employing the Jiminy co-pilot, an internally hosted Visual Studio Code Extension agent embedded in our software development environment. The Jiminy co-pilot accelerates core software development operations such as code autocompletion, unit test generation, code translation from one language to another, documentation generation, code optimization, and bug detection. Copyright risks are mitigated by, first, tuning LLMs with Lockheed Martin IP and, second, training developers to use Jiminy not to generate algorithmic solutions but also to assess code safety and security, create software tests of developed code, improve its readability, and document it. A natural language interface answers questions about the code base or developer documentation naturally.

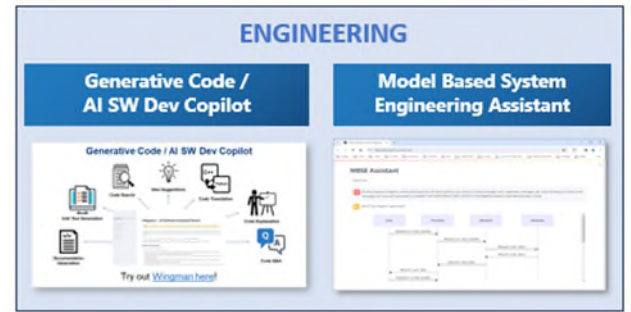


Fig. 2. Engineering use cases

In support of advanced military programs like F-22, AI Factory's low-code RAGaaS (Retrieval Augmented Generation as a Service) was used to build a chatbot that provides a user-friendly interface to access Joint Simulation Environment (JSE) information. The JSE, led by the Naval Air Warfare Center Aircraft Division (NAWCAD), is a critical component of the US military's training and testing infrastructure, enabling simulation of complex scenarios to enhance readiness and reduce costs.

The chatbot was trained on a vast corpus of data, including government documentation, online resources, code samples, and header files, and deployed on enterprise AI infrastructure. It has brought significant value to the developing program in Skunk Works by providing a centralized knowledge hub that reduces

This effort supported by Lockheed Martin Corporate

time and effort to find critical information, enables faster and more informed decision-making, and improves the overall user experience. Other programs, starting with F-22, are well positioned to apply this capability quickly.

Another LLM engineering use case, shown on the right of Figure 2, extends our Model Based System Engineering (MBSE) environment with an assistant that imports existing Systems Modeling Language (SysML) models. More important, it autogenerates SysML from natural language descriptions of system requirements. The MBSE assistant also supports Q&A on SysML models. The assistant accelerates and enhances accuracy and consistency of generative design cycles. Given use of the AI Factory framework for deploying LLMs to cloud independent networks, the platform supports classified or “air-gapped” program deployments.

### B. Enterprise Use Cases

While engineering tools support thousands of our 60,000 engineers, the Lockheed Martin Navigator shown on the left side of Figure 3 provides safe access to multiple LLMs with over 50,000 active users (42% of employees) enabling text summarization, email drafting, document interrogation, brainstorming, Q&A, proofreading, and other generative tasks across the enterprise, amplifying knowledge-worker efficiency, speed, and effectiveness.

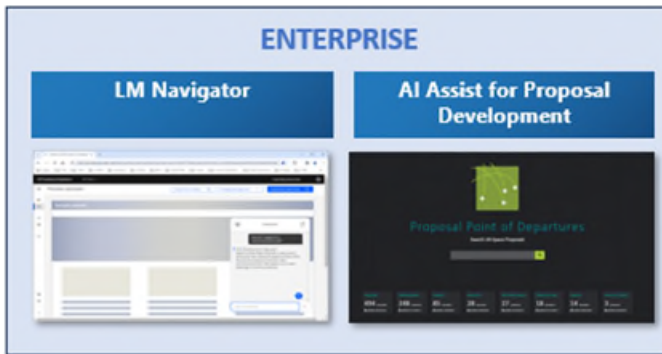


Fig. 3. Enterprise use cases

Another use case within our Space business area is illustrated on the right side of Figure 3. In this proposal development assistant, model tuning with previously successful proposals, effective prompt engineering, retrieval augmented generation (RAG), automated critique, and human review has enabled the generation of responses to over a thousand requests for proposals by generating recommendations for proposal content that helps grow sales. Proposal assistance enhances speed, accuracy, affordability, compliance, and win rates while simultaneously broadening employee knowledge.

The value of enterprise deployments such as these is enhanced by tuning open source LLMs with enterprise specific knowledge captured in structured, unstructured and semi-structured documents containing information such as corporate policies and procedures, project and quarterly reports, product catalogues, supplier descriptions, customer requirements and insights, financial records, and competitive intelligence. Tuning requires careful exclusion of any classified, export control and/or sensitive information as well as appropriate user access

control of resultant LLMs to ensure appropriate privacy, security, and confidentiality. The resultant authoritative sources can greatly enhance enterprise knowledge workflows. In related research, Bloomberg uses financial data to fine tune its internal Bloomberg GPT to enable AI assisted journalism through capabilities including automating headline generation, chart summarization, table understanding, story summarization, and financial story generation (including countering adversarial attacks) [4]. AI assisted journalism provides Bloomberg with a speed and capacity advantage while emphasizing accuracy and trust.

### C. Edge Use Cases

In addition to enterprise value, AI Factory has accelerated many missions for Lockheed Martin’s customers. MLOps is critical to supporting missions in denied or disconnect environments, where models need to be trained at the edge to meet rapidly changing mission environments. AI Factory ensures teams of engineers and operators can securely train at the edge to meet the needs of a given mission. AI Factory provides the warfighter with an automated, repeatable, and robust end-to-end machine learning pipeline and the ability to confidently retrain and deploy new models at the speed of relevance. Although not GenAI, robust MLOps processes were used in a recent example shown on the left of Figure 4 in which the U.S. Navy and Lockheed Martin developed and fielded software updates for destroyers shooting down Houthi missiles and drones in the Red Sea [5].

More specially, in collaboration with the USN, LM developed an Aegis Speed to Capability process that allows for small changes to be rapidly fielded, instead of waiting to incorporate them into the next major baseline upgrade to the combat system software, reducing the time to produce updates from months to days, and eventually hours. The data feeds LM’s AI Factory machine learning operations (MLOps) pipeline, which is used to automate training of new machine learning models for use in both offline analysis and online deployment. Analogous to the Continuous Integration / Continuous Deployment (CI/CD) pipelines used to automate software update and testing, artifacts are produced for peer review of results by U.S. Navy and LM technical experts, enabling rapid assessment of performance improvements each update will yield before graduation to at-sea testing.



Fig. 4. Edge use cases

Within the aerospace industry, as shown in the right of Figure 4, AI Factory was employed to develop an F-35 Action Request Submittal Assistant. Customer service was enhanced by

supporting 100Ks action requests for F-35 sustainment activities, automatically generating recommendations based on historical precedence.

### III. AI FACTORY ML OPERATIONS PLATFORM

Designed for scaling, security and flexibility, the Lockheed Martin AI Factory and its commercial release, the Astris Genesis™ (AstrisAI.com) platform, is a self-hosted platform architecture to enable thousands of AI/ML engineers at Lockheed Martin to operate safely and efficiently, driving critical design elements. More specifically, the AI Factory is a scalable, secure, and flexible end-to-end ecosystem designed to support the full lifecycle of artificial intelligence (AI) and machine learning (ML) models. This modular architecture enables AI teams to build, train, deploy, and sustain AI solutions efficiently, while ensuring automation, monitoring, and security at all phases of the deep learning lifecycle. By codifying AI best practices into reusable components, pipelines, and playbooks, AI Factory enhances developer productivity and facilitates the sharing of expertise across the enterprise.

The Lockheed Martin AI Factory is a Kubernetes-based ecosystem that leverages open architecture design to scale operationally at low cost across a diverse enterprise, avoiding commercial vendor lock and monolithic software solutions. This approach enables the efficient implementation of a modern AI/ML tech stack through collaboration with commercial tech companies and small to medium sized businesses. By applying MLOps and AI DevSecOps practices, AI Factory streamlines the development and operational utility of models through infrastructure, data, and machine learning pipelines.

The AI Factory is guided by the following design principles:

- *Scalability*: Scale across Lockheed Martin business areas with minimal cost to adopting programs.
- *Infrastructure Agnosticism*: Remain agnostic to data storage and compute infrastructure.
- *Modularity*: Utilize only the environments, tools, and components that are valuable to each program.
- *Comprehensive Lifecycle Support*: Cover the entire AI/ML lifecycle and inject capabilities unique to Lockheed Martin's mission.

By adhering to these principles, AI Factory provides a reusable and scalable solution for the full AI/ML lifecycle, enabling Lockheed Martin to efficiently develop and productionize AI solutions at scale

Figure 5 describes the model creation pipeline of the Astris Genesis platform steps of which are described in turn. These are designed to support not only different stages in the life cycle but also different types of users including designers, developers, and managers as well as different types of end users from functional experts (e.g., finance, legal, IT, HR, cybersecurity) to program managers to mission operators to executives.

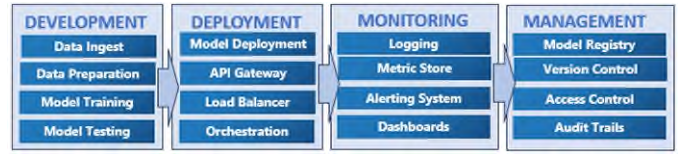


Fig. 5. AstrisAI Genesis™ AI Factory pipeline

#### A. Model Development

Model development supports the ingestion and cleansing of data enabling model training and testing. This Integrated Development Environment (IDE) helps data scientists rapidly and safely build, train, test and deploy AI models using popular frameworks such as TensorFlow and PyTorch.

#### B. Model Deployment

Automated deployment of models to production environments is supported by containerization and orchestration using Kubernetes.

#### C. Model Monitoring

Real-time monitoring and logging of model performance feeds into a metric store which drives alerts and notifications for anomalies which can be displayed in intuitive dashboards.

#### D. Model Management

Centralized management of models, data, and infrastructure, enables services including model registration, version control, access control, auditing and collaboration.

#### E. Distributed Processing

A differentiating design feature illustrated in Figure 6 is the transformation from centralized to decentralized AI. In contrast to conventional machine learning pipelines which require data to be pushed to a centralized data pool, data and models are distributed to edge nodes. This Astris Genesis distributed solution enables AI processing to occur at edge nodes. This architecture is motivated by real-world operational needs driving a design philosophy of not bringing the data to the AI but rather bringing the AI to the data. This is essential in cases such as remote critical infrastructure, driver safety in autonomous vehicles, secure/classified air-gapped environments, and remote operating locations.

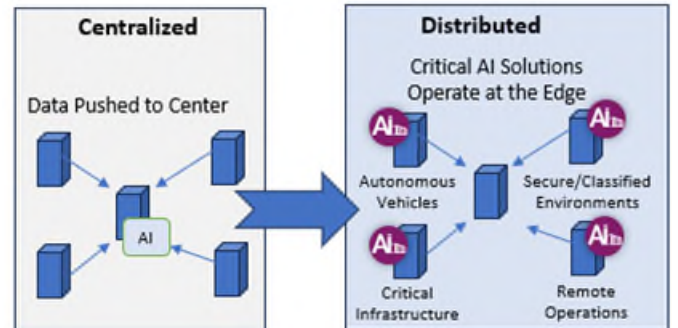


Fig. 6. Astris Genesis™ distributed AI Factory

The AI Factory MLOps platform supports the secure, efficient MLOps pipeline shown in Figure 5. This enables rapid retraining and deployment of a wide range of AI models—covering computer vision, machine learning, and advanced large



language models—right at the edge. By leveraging Kubernetes and Red Hat Device Edge, this enables:

- Rapid model updates without requiring a network connection
- Safe and secure data handling, model retraining, and over-the-air updates, essential for real-time use cases
- Flexible deployment through containerization and orchestration
- A modular, open-source design that facilitates adaptation, customization, and scaling

#### IV. ASTRIS GENESIS™ GENERATIVE AI PLATFORM

As illustrated in Figure 7, the Astris Genesis generative AI platform is built on top of the AI Factory MLOps platform which in turn is supported by processing and storage infrastructure. The Genesis AI platform offers engineers and developers a low code/no-code self-hosted, on-premises solution for generative artificial intelligence, enabling organizations to enhance innovation, streamline operations, and drive digital transformation while adhering to stringent security, compliance, and data governance standards.

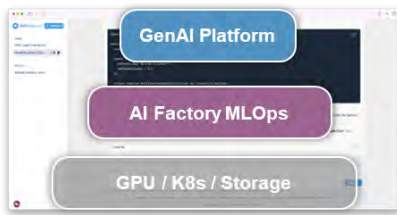


Fig. 7. Astris Genesis™ Technology Stack

In contrast to cloud-based AI solutions, the Astris Genesis platform provides a distinct advantage in terms of security, control, and flexibility. By hosting the platform on-premises, organizations can maintain full control over their data and applications, tailoring them to meet specific requirements. This approach ensures the protection of sensitive data and intellectual property, providing a secure foundation for various applications, including business process automation, insight generation, and the development of innovative products and services.

The Astris Genesis platform's architecture is designed to provide a reliable and secure environment for AI-driven innovation, allowing organizations to leverage the benefits of generative AI while minimizing the risks associated with cloud-based solutions. By choosing an on-premises deployment, organizations can ensure that their sensitive data remains within their control, reducing the risk of data breaches and unauthorized access.

The self-hosted platform architecture was created to enable thousands of AI/ML engineers at Lockheed Martin to operate efficiently, driving critical design elements. These include:

- *Best-in-class Open Source Models*: utilization of the most current, leading open source models ensures user access to the highest levels of performance and innovation

- *Safety*: Enterprise access to internally hosted leading foundational models eliminates need for off-premise access
- *Affordability*: leverage of open source models reduces costs associated with proprietary model use such as per-token and licensing fees
- *Secure by Design*: designed with data security and compliance in mind, minimizing the need for sensitive data sharing with third parties.
- *Flexible and Scalable*: built to handle large-scale datasets and complex workflows, making it an ideal solution for organizations of all sizes
- *Ease of Integration*: integrates seamlessly with existing infrastructure and workflows, minimizing disruption and downtime

#### V. LM NAVIGATOR

Safe, accessible and intuitive access to generative AI is essential to foster adoption, improve business operations and drive innovation for a broad set of customers. LM Navigator has been applied across a range of use cases including generating and testing software code, accelerating post-mission analytics, and extracting answers from extensive production line documentation [6]. Its creation was a partnership between the Lockheed Martin's AI Center (LAIC) and the corporation's ILMX digital transformation team. Careful attention to data access and protection ensures that generative AI models have appropriate access and access control to empower thousands of engineering users to deliver automated and repeatable AI capabilities into high assurance applications. Given the dynamic nature of AI across a multidomain and multinational enterprise, Lockheed Martin's use of a "hub and spoke" talent development model motivates LM Navigator to deliver the latest LLMs from a centralized, secure platform across a diverse set of globally distributed engineers and disciplines.

#### VI. RESPONSIBLE SCALING

Hardened by battlefield requirements, AI Factory was designed to meet the high performance, high security needs of highly regulated industries (e.g., finance, healthcare, energy), dynamic operations (e.g., technology, manufacturing, customer service), and contested missions (e.g., defense, critical infrastructure). Resilient deployment at scale in a large, global enterprise requires a responsible AI strategy that holistically considers governance via policy, people and technology.

##### A. Responsible AI Principles

Lockheed Martin employs ethical principles [7] across the entire AI life cycle through *responsible* design, development, deployment and use monitoring, *equity* by mitigating unintended biases in development and training and use [8], *traceability* by ensuring clear specifications, training and explanation, *transparency* by providing clear insights into AI system operations and decision-making processes for stakeholders, *reliability* through testing, validating and verifying across the life cycle, and *governability* by ensuring quality performance, supporting oversight, risk management, failure mode analysis, and, if needed, deactivation [7]. An Artificial Intelligence Ethics Advisory Committee (AI EAC) both at the

board level and across major businesses ensures implementation of these principles. While it is impossible to eliminate all risks in the face of active adversaries, persistent technology change and zero day vulnerabilities, collectively, these principles foster a culture to enhance AI safety, security and assurance.

### *B. Governance*

Responsible governance emphasizes continuous risk management that prioritizes use cases that create high value with nominal risks, residuals of which are mitigated with countermeasures. Early identification of a broad set of potential risks (e.g., financial, privacy, IP, reputation) and associated harms enables the creation of policy, technical, operational and training mitigations. For example, on premise access to open source models ensures no public release of personal, proprietary or otherwise harmful information during model tuning, prompt engineering, or response generation. It simultaneously discourages risky access to off premise sources and services.

Continuous threat monitoring and mitigation is essential given rapidly advancing generative AI technology and adversarial attacks. Controls can be applied across the life cycle including training, tuning, or inference/use stages. This centralization can ease monitoring, model drift, as well as performance impacts influenced by changes in operational or environment conditions. Centralized control will become more complicated with the emergence of distributed, multi-agent systems.

### *C. Partnerships*

The Lockheed Martin AI Factory MLOps ecosystem and Astris Genesis Generative AI platform have been designed to leverage a diverse ecosystem of technologies from leading big tech companies, commercial off-the-shelf (COTS) software providers, and open-source communities. With a total investment of over \$132 billion in AI software from commercial companies in our tech stack, we have established direct partnerships with industry leaders including Meta, HPE, Microsoft, and NVIDIA to drive innovation and collaboration. Our platform integrates a combination of open-source and proprietary software, ensuring seamless integration with industry-standard tools and frameworks. Lockheed Martin has also invested tens of millions of dollars in software development for AI Factory, encompassing the MLOps platform, Generative AI Platform, and Enterprise Chatbot. Furthermore, our internal AI Consulting services plays a critical role in evaluating and integrating new technologies from our partnership intake pipeline, which includes many smaller engineering partners and commercial tech companies. By embracing an open and collaborative ecosystem, we have created a future-proof platform that can be easily customized and extended to meet the unique needs of Lockheed Martin's customers and partners. This modular and extensible approach allows the Lockheed Martin AI Factory software and the Astris Genesis platform to scale to edge and classified environments, while providing the flexibility to adapt to evolving AI workloads and emerging technologies.

### *D. Data Protection*

Given the centrality of data to model creation and evolution, emphasis is placed on data governance including availability, quality, provenance and protection. This includes a focus not

only on training data but also on model versioning, access control, and logging/monitoring. Attention focuses on protection of personally identifiable information, protected health information, and intellectual property, including copyrighted as well as export controlled and third party information.

### *E. Infrastructure Resilience*

As deployments scale, careful architecture is essential to ensure affordability, security and resilience. For enterprise deployments such as LM Navigator supporting 50K users, on premise access to, tuning of, and integration of open source models are supported by centralized compute including a NVIDIA DGX SuperPOD [9]. This centralization delivers faster and more cost effective service than by external token consumption and also supports better use prediction enabling long term financial and operational planning, including ensuring access to constrained AI microprocessor supply. Embedded teams across the corporation are supported by a centralized development and support team in the Lockheed Martin AI Center (LAIC). Decentralized deployments are enabled by bringing AI Factory to the data at the edge to enable efficiency, security, and resilience. Centralization also helps accelerate a diversity of operational missions examples of which range from lunar and mars space travel to better detecting, predicting and fighting wildfires [9].

### *F. Talent*

Generative AI literacy is expected by all employees, even if only occasional users. Mandatory training on appropriate uses and risk management is necessary prior to use of LM Navigator or AI Factory and helps drive employee upskilling and reskilling. New use cases and development plans are reviewed prior to development and deployment by the AI EAC to both drive enterprise value and mitigate risk.

Given the criticality of AI engineering and functional expertise, as part of Lockheed Martin's 21<sup>st</sup> Century Security (21CS) vision, we cultivate AI talent via AI Learning Pathways, curated online self-paced learning materials which are linked to Lockheed Martin's AI role competencies. Together with live instructor led training, these enable staff to acquire digital badges, credentials earned after completing a set of expert-vetted learning experiences that demonstrate proficiency in critical skills. For example our hands-on AI/ML Fundamentals Program includes a cohort of 50 students who enter the program at appropriate levels based on their objectively assessed skill level. They must complete an applied AI capstone project working with a qualified mentor. Customized learning programs are available for advanced upskilling and continuing education, including unlimited access to multiple third party learning platforms (e.g., DataCamp).

## VII. RISKS AND MITIGATION

Enterprise success requires a wholistic strategy that accelerates deployment by incorporating cross disciplinary risk mitigation up front. Just as its important to employ policy, technology and human controls across the ML operations life cycle shown in Figure 5, so too it is important to anticipate intentional attacks on or unintentional failures in confidentiality,

integrity and availability [10]. These classes of failures can result in breeches, errors, and denial of service.

In addition to these traditional threats directly against LLMs, generative AI can be employed by adversaries to plan and execute novel attacks at scale. For example, voice and video deep fakes have already been employed successfully by adversaries [11] and high quality social media attacks at scale are feasible. Human users regularly fail to distinguish high quality audio and video deepfakes. Even inexpensively spoofed text chat conversations now elude human detection. In a five minute evaluation of historical chat bot psychoanalyst Eliza, it could only trick 22% of 500 test users that it was human. In the same test, humans assessed other humans as human only 66% of the time. However, those same 500 users believed GPT-4 was human 54% of the time [12]. In this study, the most effective countermeasures to human detection were direct accusation, logic and math, human experience and current event questions.

In another case, LLMs were trained to identify cyber vulnerabilities and generate exploits in software. This strategy should unfortunately increase the frequency of zero day exploits. However, just as important, these same methods can be applied by defenders to counter these threats. For example, defenders can automate the detection and patching of software vulnerabilities during design or test prior to release and, similarly, in real time or proactively search for and detect deep fakes across various media [13].

Because of the rapid development of LLMs, AI/ML risk frameworks are new and require continuous refinement. For example, building upon the Adversarial Tactics Techniques and Common Knowledge (ATT&CK®) framework [14], MITRE's Adversarial Threat Landscape for AI Systems (ATLAS™) [15] catalogues AI attack tactics, techniques and mitigations as well as case studies of previous attacks. Both are essential knowledge sources to inform threat analysts, inspire table top cybersecurity training, or guide penetration tests and/or red teams. In addition, the NIST AI Risk Management Framework [16] building upon the prior NIST RMF, identifies a broad set of AI risks and methods for measuring and controlling them.

However, it is not yet clear how to prioritize limited mitigation resources against various types of risk. Accordingly, all levels of the organization, from practitioners up to the board, and all business areas and functions (e.g., finance, IT, HR) need to collaborate to mitigate the risks of AI and GenAI while delivering its value. Organizations can mitigate threats and vulnerabilities through responsible AI policy (e.g., guidelines for acquisition, development and use of data and models, privacy and IP policies), technology (e.g., prompt and generation guardrails, bias detection and mitigation, grounding knowledge to limit hallucination, marking and tracking of generated IP), operations (e.g., on premise deployment for sensitive data and models), and people (e.g., training for responsible use as well as deep fake detection, red teams, penetration attacks). Related research identifies common attack surfaces across the life cycle of model development, tuning, and inference/use and details threats against integrity, confidentiality, and availability. It considers attacks against generative AI as well as adversarial use of generative AI to formulate and implement attacks, and offers a diversity of

methods to counter the weaknesses of biased, brittle, and baroque foundational models [10].

## FUTURE DIRECTIONS

Ensuring the sustained realization of value from LLMs will require rapid deployment of future generations of foundational models while simultaneously mitigating evolving attack surfaces. Accordingly, the Astris Genesis and AI Factory solutions will evolve through continuous assessment and integration of rapidly evolving foundational LLMs. Moreover, the platform embodies design aspects that help minimize LLM vulnerabilities and reduce potential harms.

Second, new directions in LLMs will be embraced including emerging agents who use plans and actions to support and advance the intentions and goals of human users. Early work on communicative actions [17] can be leveraged to support intentional agents that achieve outcomes through carefully selected actions, applicable into well-defined preconditions with anticipated post conditions and disablements. Related, Lockheed Martin's advancement of a Cognitive Mission Manager (CMM) combines real time sensor data with course of action planning and model based digital twins to decrease response time and increase human effectiveness in life critical missions such as wildfire fighting [18].

Building on these advancements, the Astris Genesis platform and the Lockheed Martin AI Factory tool will incorporate agentic architectures that redefine traditional approaches to AI deployment. These architectures, fundamentally modular in design, enable the creation of reusable generative AI agents that can be rapidly adapted and orchestrated to perform a wide range of tasks. Specialized *orchestration agents* will coordinate workflows across multiple stages of a process, ensuring seamless execution, while *communicator agents* will facilitate the sharing of information among agents, enabling effective collaboration. *Task-oriented agents*, such as planner and research agents, will focus on specific functions, such as mapping processes or gathering and synthesizing information to support decision making.

To ensure continuous improvement, the Astris Genesis platform and the Lockheed Martin AI Factory software will implement robust feedback mechanisms that refine agent performance, enhance collaboration, and streamline task orchestration. For example, a fleet of generative AI agents could interact with inventory, supply chain, and analytics systems to autonomously monitor stock levels, identify low inventory, and generate purchase orders - eliminating the need for complex integrations and significantly improving operational efficiency.

With the rise of agentic architectures, distributed agents will play a critical role in planning, coordinating, and executing tasks collaboratively. Ensuring trust between humans and machines will remain paramount, requiring robust mechanisms for validating agent competencies and understanding the consequences of delegated authority. Related, the growth of cooperative robotics and humanoid robots not only in manufacturing but also in broader society (e.g., delivery, healthcare, companionship) will introduce new opportunities and challenges. Accordingly, we should anticipate attacks on AI

trust and the need for improved understanding of human and agent competencies and trustworthiness.

In conclusion, realizing the full potential value for generative AI use cases will require increasing the quality and depth of representation and reasoning coupled with improvements in policy, technology, and talent development to help ensure our countermeasures keep pace with increasingly sophisticated and motivated threat actors.

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